

Alex Kiefer: "Resilience from an active inference perspective"

25:23

all right keeping with the co-authorship
theme next talk is a 25 minute recorded
talk by Alex kefir resilience and active
inference

2024 this is Alex

kefir this is Alex kefir I'm here to
present some ideas on on resilience and
active inference um so this is largely
just a unpacking in a careful way of
some Concepts from this paper from 2022
that I was involved in um but also will
explore some some novel directions um
that this uh this thinking has taken
since

then so the concept of resilience really
if you look at the etymology it's just it
means like the capacity to recoil
essentially or to or to to bounce back
or just to survive through hardship and
change um so obviously this concept is
sort of a fundamental one if you're
looking at uh life life forms um
dynamical systems more generally um in
the context of like um ecosystems and
sustainability it's an important concept
um so that's why we're interested in
this um it's actually also sort of a
fundamental Concept in Material Science
so so since this paper has come out I've
been approached a couple of times by by
journals of Material Science for um for
follow-ups and such um so as I said the
the fundamental concept has to do with
recoiling or bouncing back um but
there's sort of there are some

peripheral sort of different facets of this of this meaning and we explored all of these in this paper um so roughly speaking you can think of resilience in sort of three ways um so first there's inertia um so inertia I mean I guess literally just means really uh uh sort of not changing um so persistence uh but you can also think of it as resistance in the sense that right unless you're unless you're a exist in a vacuum uh then you're a system that's embedded in some broader system and inertia is basically continuing on your way despite changes uh that might that might occur or interactions with that broader system um so if you think of the red arrow as representing the path of a of a subsystem or an H through its state space then and The Black Arrow is some kind of potential disturbance um then the top diagram here illustrates inertia um the central meaning again of resilience arguably is elasticity or the ability to bounce back so in that case you undergo some perturbation or change um and you end up recovering your initial doesn't have to be that you recover your exact initial position but the point is you make your way back to a reasonable state in terms of the kind of system that you are um then there's also this sort of third thing called we call plasticity or adaptation um so in this case you begin in one configuration uh the red arrow at the beginning here and after the perturb uh the perturbation or disturbance you

end up in a slightly different configuration so that should be read not as just that you change your state like you update your um your beliefs in the time scale of inferences or something but you actually you actually wind up somewhat adjusting your model in response to this change um so this is more like something like Allostasis um whereas plasticity is returning to the same state um so I say there are some close relationships just to get into some philosophical conceptual analysis for a second here um the first thing uh inertia probably doesn't really exist in its pure form right in nature in a way or it if it does it reduces to sort of elasticity so you know I mean presumably if you if you perturb a system um then you have to cause some change in the system however slight um otherwise there hasn't been a physical interaction right uh and inertia is going to look in practice just like the capacity to sort of instantly recover um from that tiny from that tiny change um there there is in the limit case there is sort of a different meaning here because again there might just not be a change but I think um I think in the context of realistic systems there will always be some some interaction um but arguably these two these two concepts are very close um plasticity seems like um on the face of it like a different thing because um well first of all uh the ability to be to be plastic in this

sense sort of explains how you could come to have this property of elasticity um so if you look at the example of neuroplasticity for example um it's the ability to adapt to change and to basically update your your synaptic weights to arrive at a new model that allows you to be uh in part allows you to be resilient in these first two senses um but there's another sense in which if you just assume that there's some property of the system that doesn't change through this um through this process which I think is a reasonable assumption in the case of uh you know biological systems um then there's going to be some kernel of the system that survives through change and then you're back to your sort of original definition um okay so if we look at this concept through the lens of the free energy principle um I mean first of all it turns out to be a very fundamental at you know as is often the case when you try to unpack um sort of uh fundamental properties of of systems in nature in terms of the FP and active inference uh you find that there's a fairly direct mapping from the properties that the FP attributes to systems or um the types of systems that are well described by the FP and this uh and this concept of resilience um so I mean Carl Friston has said in discussions of this that essentially uh systems well described by the fdp are resilient um uh full stop so uh but we can look at at the three uh facets of

the meaning of resilience here so
um certainly systems described by the Fe
are they have this property of inertia
in the sense that um by definition right
they're things that have managed to
persist so they limit their exposure to
uh to unlikely States unlikely
observations um but they're also sort of
by definition elastic systems um because
because they have this attracting set so
um having an attracting set of States
means that um you know of course there
might be changes or perturbations that
are sort of catastrophic that destroy
the system but as long as the system is
not destroyed then by definition it's
going to make it make its way back to
the subtracting set of States um so
resilience in the core sense is really a
um almost a defining feature of systems
that are described by the FP uh and then
finally plasticity you can think of as
something like basian model selection um
on some time scale and so that that
might seem like something extra but it's
only extra if the multiscale aspect of
the FP is extra and I would argue that
that it really isn't because at its
heart the FP is a is a scale-free
description of things and this
multiscale and scale free are sort of uh
sort of um connected Notions um so
that's
plasticity um okay but so most of the
interest I'd say of this lies in the the
mapping of resilience onto this this
formalism um of active inference so uh a
key concept uh in this analysis is the

notion of degeneracy and this is is a uh somewhat technical term um here we're follow we mainly follow on the definition of and and in fact the analysis of degeneracy in terms of active inference in uh Sajid at all 2020 this is a great paper on this topic um so here we have just the expression for variational free energy um in the top row um is a expression in terms of basically a Helm hols free energy so it's an energy term minus an entropy term uh uh so if you just look at that for a second um the we've identified uh again following this paper from 2020 we've identified degeneracy with the entropy term um so we'll get into why that is in a second um but basically uh if the concept of degeneracy is useful redundancy um so this you can think of this as like complexity that that earns its keep or that's worth having um so complexity in this this bottom breakdown of variational free energy uh you uh you analyze it in terms of complexity which is a the the difference the K Divergence between the prior belief uh Over States the the generative density Over States and the approximate posterior so this is kind of like how how difficult is it or how much change has to uh does the system have to undergo in order to represent the states that are um inferred from some observations um what's what's what's the departure between that posterior distribution and the prior dist distribution Over States

uh that's complexity um minus accuracy
um the the the expected uh log
probability of observations given
States um so what's the connection
between complexity and this idea of
useful redundancy uh sorry of just
redundancy um there's there's a
assumption here that complexity and
redundancy are closely related um I
think the basic idea is that so
redundancy uh in this context is quite
literally just duplication of States um
so the I'd say the classic example is
like an error correcting code so if
you're trying to transmit some uh a bit
of information over a channel uh you
might you might send 10 copies of a bit
in case in case some of those bits get
flipped or corrupted uh and you want to
make sure that the that the message gets
transmitted um so that's redundancy it
increases the complexity because um uh
well one way of looking at it is that it
uh it it it increases the description
length uh right this this K Divergence
is like how how how much how much more
do you have to say in order to
communicate that an event drawn from
this Q distribution has occurred um G
given that you assume this P
distribution um so uh so you can think
of it as well what these these extra
bits are are the Redundant bits um so
there there's a sort of conceptual
connection between complexity and
redundancy
um and then degeneracy on this first
definition is just the complexity that

serves some purpose and we'll get into that in the moment um it's just also important to note that there's this definition of um degeneracy as partial redundancy in the in the literature uh and this is partial overlap and function so um the idea is that um redundancy exists in a system uh sorry degeneracy resists uh exists to the extent that multiple states in the system serve some function in common but then those States also have um otherwise uh have distinct or disjoint sets of of functions um and in practice I think these two almost always line up so there might be sort of uh limit edge cases in which um in which they're different but in but in practice um right if you have multiple States they'll always have some different properties and therefore um perform some different functions

okay so so this is one of the core things that we pointed out in this paper um of ours so um this this actually sorry this breakdown is from the sjad all 2020 paper uh but we sort of have a different gloss on the Central Center term here um so this is the complexity term in the variational free energy you can break this down in terms of um further break this down in terms of this uh expectation under Q of the negative log probability of States under the generative model um so this is called a cost term in the 2020 paper um we describe this in epistemic terms as as an uncertainty term so this is uh

this is basically uh you can you can read this as the Shannon uh minimum description length of the of the states given the generative density um but it's sort of the expected surprise um how surprising are are these states that I that I infer uh using the variational posterior under my generative model um so that's an uncertainty term uh and then we subtract this degeneracy or the entropy over um over the variation of posterior Over States um so I mean first of all sorry I realized that I didn't quite make the the link here between um degeneracy and entropy so why is this the case uh why do we identify these uh it's it's because um this means if if the if if this Q distribution Over States is higher in entropy it means that there's a um degenerate mapping in the likelihood from states to observations right so meaning for multiple States uh you can get the same observation um and so then when you invert that likelihood uh mapping in order to recover this posterior um this means that for one observation you might have several um several different states and the spread of the of the distribution Over States of course is just the entropy so that's that's the link to to genery there uh so so this complexity term is uncertainty minus degeneracy um one reason that's interesting is that when this complexity is zero um it's not as if the system is completely certain of what's going on um

the uncertainty term actually remains
it's just balanced by this degeneracy in
other words all of the uncertainty in
this ideal case um is
is useful um and so I think this really
this really captures what what
degeneracy in this context amounts to um
it's uncertainty that's useful and is
it's useful precisely because there's
some amount of uncertainty that's
inherent in the source of the
information you're trying to model right
so um here p is the the in this context
is the generative model that the system
has of um it's the sort of internal
generative model of of the external
States um and so there will be some
right there'll be some entropy some
Source entropy associated with that
system that that you're trying to model
and um so long as you match your
variational posterior matches that
degree of uncertainty um then you're in
sort of an optimal um an optimal model
so in other words for maximum model
goodness try to ensure that um P equals
 Q um so uh this ties nicely into the
idea plasticity resilience is plasticity
so your goal here is to adjust the
posterior to match the generative model
essentially um so you can do that um you
know in in like perceptual inference uh
you can try to infer a state uh a q
distribution this is again an
approximation to the posterior the true
posterior over over those States under
the generative model um but equally you
can adjust the generative model in in a

in a multiscale setting you can learn
learn or adjust update the parameters of
the generative density in order to match
what's in the external World um
so right this this complexity term um is
affected by changes to p as well as
changes in Q and so I think the
fundamental idea that I had here was
that
um in an ideal uh sort of optimal system
under the free energy principle you
should expect um first of all P will be
adjusted so as to model the environment
for sort of um basic you know um free
energy principle SLG good regulator
theorem types of reasons uh and then Q
will also be adjusted to match to match
 P um so this involves um this cybernetic
principle called The Law of requisite
variety which is sort of a um um it's
closely related to the good regul good
regulator theorem um but basically it
just says that you know if you're trying
to regulate or control some States then
you need uh at least as much uh variety
or uh in probabilistic terms that that
translates to variance um or or entropy
um you need as much Variety in the
controller as you have in the in the
environment and so it's a fairly
intuitive thing because you basically
need sufficiently many degrees of
freedom in the controller uh to be able
to map to to model what's in the
environment um so you can see this as
this degeneracy um
property of resilient systems or uh any
system under the FP as as an instance of

this law of requisite variety it's also clearly by the way a an obvious case of constrained maximum entropy so there's work um showing that many many lines of thought really showing that the active inference uh sorry the free energy principle is um is equivalent to a principle of constrained maximum entropy uh so here the constraint is this uh you can you can think of this as constrained by accuracy um but you you basically maximize this entropy term subject to that constraint uh subject to the generative model okay so um looking just a bit more closely at this notion of degeneracy um so I've said that it's Central um one pointed question you might ask here is well are all do all resilient systems uh have this degeneracy property uh and so I want to say the answer is yes um so the type of degeneracy that we've already talked about is this degeneracy and the likelihood mapping uh so this is multiple States mapping to the same observation um and again that's uh that can be cast as you'd expect to see that given um that you're performing maximum entropy inference um so that that is um crucial but equally important I'd say is degeneracy and the Dynamics uh so this characterizes just systems that again that have an attracting set of States so if these STS are states at a given time and then st+ one is the state or set of States at the next time step um so this in order for for a system to be resilient um it

really has to be the case that this transition um uh that these transition probabilities or the Dynamics of the system are such that um that that mapping itself is degenerate um so you know this means that basically you can perturb the system in several different ways and you'll still arrive uh you'll still end up at the same at the same uh state or or set of attracting States um obviously if the if the perturbation is too large then that might not be the case the system will diverge and you'll destroy you'll destroy it but by definition the systems that survive through change are the ones that have this property um so there is a very deep link between resilience and generality if you consider Dynamics as well um just a couple more points here one is um something that I've recently been thinking about uh it's it's it's actually a theme that's been on my mind for some time but I've recently written about it in some other work um and this is what what does the principle of requisite variety and um what do these concepts of degeneracy and resilience look like in in the context of multiscale active inference um but specifically here I don't I don't just mean um the fact that excuse me the fact that model parameters as well as states can change but I mean U multiscale active inference in the sense that you've got potentially higher level agents that are composed of lower level

agents um so in that kind of setting uh
requisite variety ends up looking like
the the property of higher level agents
that they're composed of a variety of
different types of lower level agents um
so I mean I would argue that if you
combine this law of requisite variety
with multiscale active inference in this
sense then you really sort of
automatically have something like a a
diverse ecosystem in which there are
many different types of agents that
participate um so here in this diagram
we're illustrating first of all just the
same degenerate State observation
mapping that we saw in the uh in the
previous slides um but here the nodes
are individual agents that compose this
higher level
agent
um also as pointed out in this paper
um by by um Carl friston and Daniel
fredman at all um the in this kind of
context different agents can be expected
to and I think empirically in some
simulations are shown to take on the
roles of these different types of States
so uh of course the breakdown here is in
terms of active and sensory and uh
active sensory and internal States so
here you've got active sensory and
internal agents uh and that's one I
would argue of this um application of
the of the law of requisite
variety um to this higher level uh
system um I also show here the the
policy selection based on expected free
energy for completeness um but that's

sort of the main the main point is this
this specialization and diversification
of Agents um in this in this setting um
in terms of the second definition of
degeneracy you can also think of these
agents as playing um somewhat distinct
but partially overlapping roles um and
there's some some lots of interesting
work on um on that sort of multi-agent
perspective from the point of view of
active

inference um so that's really it I'm
just going to conclude with some with
sort of a philosophical question that I
think is is kind of important for um for
tying this this together and and sort of
checking that it that it just makes
sense um so I've said earlier that um
any system well described by the FP as
res so hold on a second uh not all
systems are equally resilient though uh
right tups by are sort of the paradigm of
a fragile as opposed to resilient object
so since the FP describes things in
general doesn't it mean that all things
are resilient um and doesn't that
contradict this obvious fact so I think
the answer here is that it's simply it's
a matter of scale so yes all things are
resilient all things that is things that
are reidentified over time and that
persist through change um all such
things are resilient but it's a matter
of scale so if you if you introduce a
very slight perturbation to this
teacup um very slight you know um
relative to a certain a certain
reference frame or a certain scale but

large enough right relative to the scale
on which that system is resilient then
by definition it will survive right and
so obviously again in a in a a real
universe in which there are interactions
and you're you're always interacting in
some way dealing with some sort of um
random
fluctuations um then yes for the system
to persist through time to be inert and
elastic um it has some right the
material has some uh coherence that
allows it to to persist through those
changes so it is a resilient system it's
just that if you introduce a change
that's um on a scale that's That's too
great for the system to handle then
it'll be destroyed um but sort of the
just the basic coherence as a spao
temporal object of this teacup is itself
a form of resilience uh and then we get
when we get into more complex forms of
life more complex generative models
that's when we when we when we begin to
see more um impressive forms of
resilience and shading it even into into
plasticity and the ability to to really
survive through more significant
changes so thanks for for listening SL
watching um I'm happy to participate in
this Symposium um thanks to all the
other uh the co-authors of this um paper
by Mark Mall in 2022 that was the
departure point for this work uh yeah
thank you